

# Fundamental Group

## 1.1 Basic Constructions

### Definition 1.1.1 ► Product path

Let  $X$  be a topological space. Let  $\omega, \tau : \mathbb{I} \rightarrow X$  be two paths such that the terminal point  $\omega(1)$  of  $\omega$  is the same as the initial point  $\tau(0)$  of  $\tau$ . We then define  $\omega * \tau : \mathbb{I} \rightarrow X$  by the formula

$$\omega * \tau(t) = \begin{cases} \omega(2t), & 0 \leq t \leq 1/2, \\ \tau(2t - 1), & 1/2 \leq t \leq 1. \end{cases} \quad (1.1)$$

### Lemma 1.1.2

If  $\omega_1 \simeq \omega_2$  and  $\tau_1 \simeq \tau_2$  and  $\omega_1 * \tau_1$  is defined then  $\omega_2 * \tau_2$  is defined and we have,  $\omega_1 * \tau_1 \simeq \omega_2 * \tau_2$ .

### Definition 1.1.3

We can use the standard parameterisation of any closed interval  $[a, b]$  by the interval  $[0, 1]$  and think of a map  $\gamma : [a, b] \rightarrow X$  as a path in  $X$ , viz.,

$$\hat{\gamma}(t) := \gamma((b - a)t + a).$$

### Theorem 1.1.4 ► Invariance under subdivision

Let  $\gamma : [0, 1] \rightarrow X$  be a path, and  $0 < t_1 < \dots < t_n < 1$ . Then  $\gamma$  is path homotopic to  $\widehat{\gamma|_{[0, t_1]}} * \dots * \widehat{\gamma|_{[t_n, 1]}}$ .

### Definition 1.1.5 ► inverse path

For a path  $f$  from  $x_0$  to  $x_1$ , the **inver path**  $\bar{f}$  from  $x_1$  back to  $x_0$  is defined by

$$\bar{f}(s) = f(1 - s)$$

**Proposition 1.1.6**

Let  $f$  is a path from  $x_0$  to  $x_1$ ,  $\bar{f}$  is the inverse path. Then  $f * \bar{f}$  and  $\bar{f} * f$  are null homotopy.

*Proof sketch.*

定义截断路径  $f_t(s)$ , 先沿着路径走  $(1-t)$  时间, 走到  $f(1-t)$ , 再在  $f(1-t)$  处停留  $t$  时间.

*Proof.* Define

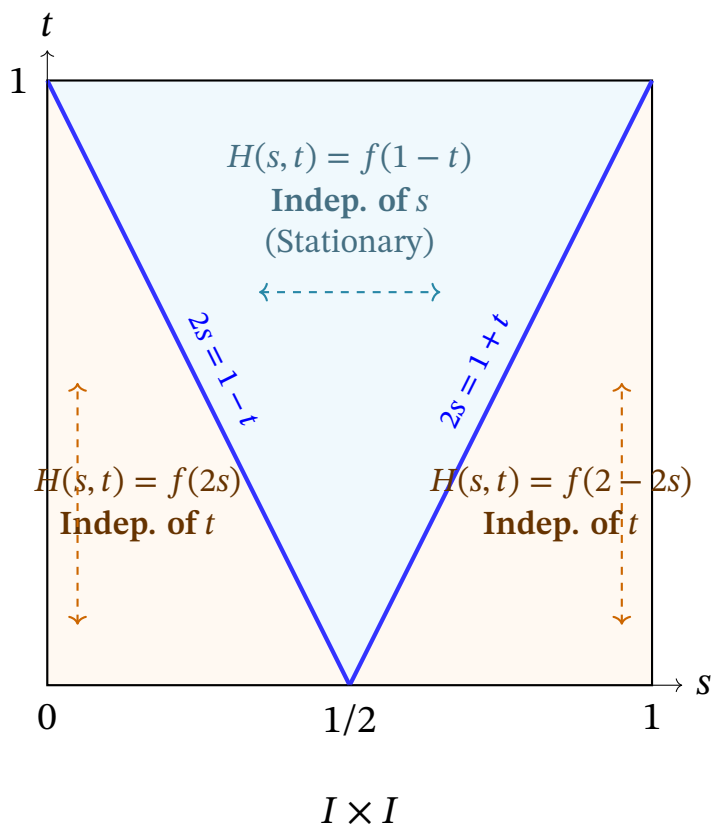
$$f_t(s) = \begin{cases} f(s), & 0 \leq s \leq 1-t \\ f(1-t), & 1-t \leq s \leq 1 \end{cases}$$

Let  $g_t = \overline{f_t}$ , we have

$$g_t(s) = f_t(1-s) = \begin{cases} f(1-t), & 0 \leq s \leq t \\ f(1-s), & t \leq s \leq 1 \end{cases}$$

Define  $h_t = f_t * g_t$ , then define  $H : I \times I \rightarrow X$ ,  $H(s, t) = h_t(s)$ .

$$\begin{aligned} H(s, t) = h_t(s) &= \begin{cases} f_t(2s), & 0 \leq s \leq \frac{1}{2} \\ g_t(2s-1), & \frac{1}{2} \leq s \leq 1 \end{cases} \\ &= \begin{cases} f(2s), & 0 \leq 2s \leq 1-t \\ f(1-t), & 1-t \leq 2s \leq 1 \\ f(1-t), & 0 \leq 2s-1 \leq t \\ f(2-2s), & t \leq 2s-1 \leq 1 \end{cases} \\ &= \begin{cases} f(2s), & 0 \leq s \leq \frac{1-t}{2} \\ f(1-t), & \frac{1-t}{2} \leq s \leq \frac{t+1}{2} \\ f(2-2s), & \frac{t+1}{2} \leq s \leq 1 \end{cases} \end{aligned}$$



$H$  is continuous on the cyan region and the two yellows region, thus is continuous.  $\square$

### Definition 1.1.7 ► The Fundamental Group

Let  $X$  be any topological space and  $x_0 \in X$ . By the above two lemmas it follows that the set  $\pi_1(X, x_0)$  consisting of all path-homotopy classes of loops in  $X$  based at  $x_0$  forms a group under the path composition. We call this group *the fundamental group* of  $X$  at  $x_0$ .

## 1.2 The Fundamental Group of the Circle

### Theorem 1.2.1

$\pi_1(S^1)$  is an infinite cyclic group generated by the homotopy class of the loop  $\omega(s) = (\cos 2\pi s, \sin 2\pi s)$  based at  $(1, 0)$ .

## Applications

### Lemma 1.2.2

$S^{n-1}$  is not the retract of the  $\mathbb{D}^n$ , that is there is no continuous map  $r : \mathbb{D}^n \rightarrow S^{n-1}$ .

*Proof sketch.*

- 借助锥构造: 收缩是恒等的延拓,  $\mathbb{D}^n$  是  $S^{n-1}$  锥, 能延拓到锥当且仅当是零伦. 由此导出恒等映射是零伦.

*Proof.* If there is such a map  $r$ , then it is an extension of  $\text{Id}_{S^{n-1}}$ , then  $\text{Id}_{S^{n-1}}$  is a null homotopy from Theorem ?? . But it follows that  $\text{Id}_{\pi_{n-1}(S^{n-1})}$  is the zero map, which is a contradiction to  $\pi_{n-1}(S^{n-1}) = \mathbb{Z}$   $\square$

*Proof sketch.*

函子性: 若收缩映射存在, 考虑

$$\pi_1(S^1) \xrightarrow{i_*} \pi_1(D^2) \xrightarrow{r_*} \pi_1(S^1)$$

而  $r_* \circ i_* = \text{Id}_{\pi_1(S^1)}$ , 但是  $\pi_1(D^2) = 0$ .

### Theorem 1.2.3 ▶ Brouwer Fixed Point Theorem

Every continuous map  $g : \mathbb{D}^n \rightarrow \mathbb{D}^n$  has a fixed point.

*Proof sketch.*

若没有不动点, 则每一对  $(x, g(x))$  确定一条射线打到  $S^{n-1}$  上的一个点  $x \rightarrow g(x) \rightarrow S^{n-1}$ , 给出连续映射  $\mathbb{D}^n \rightarrow S^{n-1}$ , 与 Lemma 1.2.2 矛盾.

### Lemma 1.2.4

There is no continuous map  $g : S^2 \rightarrow S^1$ , satisfying  $g(x) = -g(-x)$ .

*Proof sketch.*

- 存在这样的  $g$ , 意味着我们能把  $S^2$  的赤道  $\eta : I \rightarrow S^2$  通过  $g : S^2 \rightarrow S^1$  搬到  $S^1$  上,  $h = g \circ \eta : I \rightarrow S^1$ , 它是零伦的.
- $g(x) = -g(-x)$  导出  $h\left(s + \frac{1}{2}\right) = -h(s)$ , 也就是说  $h\left(s + \frac{1}{2}\right)$  和  $h(s)$

在  $S^1$  上差了半圈.

- $h$  提升到  $\tilde{h}$ ,  $\tilde{h}\left(s + \frac{1}{2}\right)$  和  $\tilde{h}(s)$  差了个半整数.
- $\tilde{h}(0)$  和  $\tilde{h}(1)$  只能差奇数
- $h$  在  $\pi_1(S^1)$  中对应到一个奇数, 与  $h$  零伦矛盾.

### Theorem 1.2.5

Every nonconstant polynomial with coefficients in  $\mathbb{C}$  has a root in  $\mathbb{C}$ .

*Proof sketch.*

- 考虑  $f(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$
- 考虑  $f$  沿着径向和切向的两族变化  $f_r(s) = f(re^{2\pi is})$ .
- 将  $f$  做范数归一化和关于  $s$  的曲线的起点归一化,

$$F_r(s) = \frac{f_r(s)/f_r(0)}{|f_r(s)/f_r(0)|}$$

- 则  $F_r(s)$  关于  $s$  是  $S^1$  上的 loop.
- 以两种方式计算  $F_r(s)$  在  $\pi_1(S^1)$  对应的元素.
- 一种是利用多项式无根,  $F_r(s)$  随着  $r$  变化到  $F_0(s) \equiv 1$ .
- 另一种考虑到, 当  $r$  充分大时, 除  $z^n$  以外的项可以看成是边缘的扰动,  $f$  沿着  $z^n + t(a_{n-1}z^{n-1} + \cdots + a_0)$  变化到  $z^n$ , 从而  $F_r(s)$  变化到  $e^{2\pi is}$ .

### Theorem 1.2.6

For every continuous map  $f : S^2 \rightarrow \mathbb{R}^2$  there exists a pair of antipodal points  $x$  and  $-x$  in  $S^2$  with  $f(x) = f(-x)$ .

*Proof sketch.*

若不存在, 则  $x \mapsto (f(x) - f(-x))$  非退化, 这是一个奇映射. 我们可以将它归一化到  $S^1$  上, 这就与上面的 Lemma 矛盾.

## 1.3 Induced homomorphisms

### Definition 1.3.1

Given a map  $f : (X, a) \longrightarrow (Y, b)$ , we have a homomorphism  $f_{\#} : \pi_1(X, a) \longrightarrow \pi_1(Y, b)$  defined by

$$f_{\#}[\omega] = [f \circ \omega]. \quad (1.2)$$

Moreover, we have the following two properties which are verified directly.

1. For any space  $X$ , if  $Id : X \longrightarrow X$  denotes the identity map then the induced homomorphism on the fundamental groups is the identity homomorphism, i.e.,  $Id_{\#} = Id$ .
2. Given maps  $f : (X, a) \longrightarrow (Y, b)$  and  $g : (Y, b) \longrightarrow (Z, c)$  we have  $(g \circ f)_{\#} = g_{\#} \circ f_{\#}$ .

*Proof.* This is well defined for if  $\omega \simeq \tau$  then  $f \circ \omega \simeq f \circ \tau$ . Since  $f \circ (\omega * \tau) = (f \circ \omega) * (f \circ \tau)$ , it follows that  $f_{\#}$  is a homomorphism.  $\square$

### Theorem 1.3.2

Let  $\tau$  be a path in a space  $X$  with  $a, b$  as its initial and terminal points, respectively. Then the assignment  $[\omega] \mapsto [\bar{\tau} * \omega * \tau]$  defines an isomorphism  $h_{[\tau]} : \pi_1(X, a) \rightarrow \pi_1(X, b)$ .

*Proof.*  $h_{[\tau]}$  is a homomorphism since

$$h_{[\tau]}[\omega_1 * \omega_2] = [\bar{\tau} * \omega_1 * \omega_2 * \tau] = [\bar{\tau} * \omega_1 * \tau * \bar{\tau} * \omega_2 * \tau] = h_{[\tau]}[\omega_1] \cdot h_{[\tau]}[\omega_2]$$

Furthermore

$$h_{[\tau]} \circ h_{[\bar{\tau}]}([\omega]) = [\tau * \bar{\tau} * \omega * \tau * \bar{\tau}] = [\omega]$$

Thus

$$h_{[\tau]} \circ h_{[\bar{\tau}]} = \text{Id}_{\pi_1(X, b)}$$

Similarly

$$h_{[\bar{\tau}]} \circ h_{[\tau]} = \text{Id}_{\pi_1(X, a)}$$

$\square$

**Proposition 1.3.3**

If a space  $X$  retracts onto a subspace  $A$ , then the homomorphism

$$i_* : \pi_1(A, x_0) \rightarrow \pi_1(X, x_0)$$

induced by the inclusion  $i : A \hookrightarrow X$  is injective. If  $A$  is a deformation retract of  $X$ , then  $i_*$  is an isomorphism.

**Proposition 1.3.4**

For a space  $X$ , the following three conditions are equivalent:

- (a) Every map  $S^1 \rightarrow X$  is null homotopic.
- (b) Every map  $S^1 \rightarrow X$  extends to a map  $D^2 \rightarrow X$ .
- (c)  $\pi_1(X, x_0) = 0$  for all  $x_0 \in X$ .

Deduce that  $X$  is simply-connected, iff all the maps  $S^1 \rightarrow X$  are homotopic.

*Proof.* The equivalence of (a), (b) has been shown in ??.

(b)  $\implies$  (c): For  $\gamma : I \rightarrow X$ ,  $\gamma(0) = \gamma(1) = x_0$ . There exists  $\tilde{\gamma} : S^1 \rightarrow X$ ,  $\tilde{\gamma} \circ \pi = \gamma$ , where  $\pi : I \rightarrow S^1$  is the quotient map, such that  $\pi(0) = \pi(1) = 1$ , as well as a map on  $D^2$  based at 1.  $\tilde{\gamma}$  extends to  $F : D^2 \rightarrow X$ , such that  $F \circ i = \tilde{\gamma}$ , where  $i : S^1 \hookrightarrow D^2$ . We have  $\gamma = F \circ i \circ \pi$ . Then

$$[\gamma] = (F \circ i)_* [\pi] = F_* \circ i_* ([\pi])$$

Since  $\pi_1(D^2) = 0$ , we have  $F_* = 0$ , thus  $[\gamma] = e_{\pi_1(X, x_0)}$ . It follows that  $\pi_1(X, x_0) = 0$ .

(c)  $\implies$  (a): Let  $\tilde{\gamma} : S^1 \rightarrow X$  be a map. Let  $\gamma = \tilde{\gamma} \circ \pi$ ,  $x_0 = \tilde{\gamma}(1)$ , then  $\gamma(0) = \gamma(1) = x_0$ .  $\gamma$  is a loop in  $X$  based at  $x_0$ . Since  $\pi_1(X, x_0) = 0$ ,  $\gamma \simeq c_{x_0}$ .  $H : I \times I \rightarrow X$ ,

$$H(x, 0) = \gamma, \quad H(x, 1) = x_0$$

Since  $H(0, t) = H(1, t) = x_0$ . There exists continuous map  $\tilde{H} : S^1 \times I \rightarrow X$  such that  $H = \tilde{H} \circ (\pi \times \text{Id}_I)$ .

$$\tilde{H}(e^{2\pi i x}, 0) = \tilde{H} \circ (\pi \times \text{Id}_I)(x, 0) = H(x, t) = \gamma(x) = \tilde{\gamma}(e^{2\pi i x})$$

Thus  $\tilde{H}(\cdot, 0) = \tilde{\gamma}$ .

$$\tilde{H}(e^{2\pi i x}, 1) = H(x, 1) = c_{x_0}$$

Thus  $\tilde{\gamma}$  is null homotopic. □

### Proposition 1.3.5

We can regard  $\pi_1(X, x_0)$  as the set of basepoint-preserving homotopy classes of maps  $(S^1, s_0) \rightarrow (X, x_0)$ . Let  $[S^1, X]$  be the set of homotopy classes of maps  $S^1 \rightarrow X$ , with no conditions on basepoints. Thus there is a natural map  $\Phi : \pi_1(X, x_0) \rightarrow [S^1, X]$  obtained by ignoring basepoints. Show that  $\Phi$  is onto if  $X$  is path-connected, and that  $\Phi([f]) = \Phi([g])$  iff  $[f]$  and  $[g]$  are conjugate in  $\pi_1(X, x_0)$ . Hence  $\Phi$  induces a one-to-one correspondence between  $[S^1, X]$  and the set of conjugacy classes in  $\pi_1(X)$ , when  $X$  is path-connected.

*Proof.* Take arbitrary  $f : S^1 \rightarrow X$ , suppose  $f(1) = x_1$ . Let  $\tau : I \rightarrow X$  such that  $\tau(0) = x_0$ ,  $\tau(1) = x_1$ . Define

$$g(e^{2\pi is}) = \begin{cases} \tau(3s), & 0 \leq s \leq \frac{1}{3} \\ f(3s-1), & \frac{1}{3} \leq s \leq \frac{2}{3} \\ \bar{\tau}(3s-2), & \frac{2}{3} \leq s \leq 1 \end{cases}$$

Since  $g$  can be given by continuously changing the basepoint of  $f$ . Then  $f$  is free homotopic to  $g$ . We have  $\Phi([g])$  is the class of  $f$  in  $[S^1, X]$ . Thus  $\Phi$  is onto.

$\Phi([f]) = \Phi([g])$  iff there exists  $H : I \times I \rightarrow X$  such that  $H(\pi(\cdot), 0) = f$ ,  $H(\pi(\cdot), 1) = g$ . By evaluating  $H$  on  $\partial(I \times I)$  counter-clockwisely, we get a loop homotopic to  $f * \tau * \bar{g} * \bar{\tau}$ , where  $\tau$  is a loop given by evaluating  $H$  on  $\{1\} \times I$  from above. But  $\pi_1(I \times I) = 0$ , we get

$$[f][\tau][\bar{g}][\bar{\tau}] = 1$$

Then

$$[f] = [\tau][g][\tau]^{-1}$$

We get  $[f]$  and  $[g]$  are conjugate in  $\pi_1(X, x_0)$ .

Conversely, if  $[f]$  and  $[g]$  are conjugate in  $\pi_1(X, x_0)$ , then  $f \simeq \tau * g * \bar{\tau}$ , by defining

$$\tau_t = \begin{cases} \tau(s), & 0 \leq s \leq 1-t \\ \tau(1-t), & 1-t \leq s \leq 1 \end{cases}$$



By defining

$$H(s, t) = (\tau_t * g * \bar{\tau}_t)(s)$$

We get  $f$  is free homotopic to  $g$ . □

## Applications

### Lemma 1.3.6

If a space  $X$  is the union of a collection of path-connected open sets  $A_\alpha$  each containing the basepoint  $x_0 \in X$  and if each intersection  $A_\alpha \cap A_\beta$  is path-connected, then every loop in  $X$  at  $x_0$  is homotopic to a product of loops each of which is contained in a single  $A_\alpha$ .

*Proof sketch.*

- 给定一个 loop  $f : I \rightarrow X$ .
- 取  $I$  的一个覆盖  $\{f^{-1}(A_\alpha)\}$  的 Lebesgue 数, 给出  $I$  的一个分划  $0 = s_0 < s_1 < \dots < s_m = 1$ , 每一段完整地落在某个  $f^{-1}(A_\alpha)$  中.
- 在每个分界点  $s_i$  处, 到  $x_0$  折返一次, 切出一段回路.

### Proposition 1.3.7

$\pi_1(S^n) = 0$ , if  $n \geq 2$ .

*Proof sketch.*

基点取成赤道上一点, 分别取南北半球多一点的片构成覆盖, 这两个片都是可缩的, 故每个道路拆成两个可缩的道路, 是零伦.

### Corollary 1.3.8

$\mathbb{R}^2$  is not homeomorphic to  $\mathbb{R}^n$  for  $n \neq 2$ .

*Proof sketch.*

- 设  $\varphi$  是同胚, 则  $\mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}^n \setminus \{\varphi(0)\}$  同胚.
- $\mathbb{R}^n \setminus \{\varphi(0)\}$  平移到  $\mathbb{R}^n \setminus \{0\}$ , 形变收缩到  $S^{n-1}$ .
- $\pi_1(S^{n-1}) = 0$  若  $n \geq 2$ .
- $\mathbb{R}^2 \setminus \{0\}$  形变收缩到  $S^1$ ,  $\pi_1(S^1) \simeq \mathbb{Z}$ .
- $\pi_1(S^{n-1}) \not\simeq \pi_1(S^1)$

## 1.4 Van Kampen's Theorem

### Theorem 1.4.1

If  $X$  is the union of path-connected open sets  $A_\alpha$  each containing the basepoint  $x_0 \in X$  and if each intersection  $A_\alpha \cap A_\beta$  is path-connected, then the homomorphism  $\Phi : *_{\alpha} \pi_1(A_\alpha) \rightarrow \pi_1(X)$  is surjective. If in addition each intersection  $A_\alpha \cap A_\beta \cap A_\gamma$  is path-connected, then the kernel of  $\Phi$  is the normal subgroup  $N$  generated by all elements of the form  $i_{\alpha\beta}(\omega)i_{\beta\alpha}(\omega)^{-1}$  for  $\omega \in \pi_1(A_\alpha \cap A_\beta)$ , and hence  $\Phi$  induces an isomorphism  $\pi_1(X) \approx *_{\alpha} \pi_1(A_\alpha)/N$ .

*Note.*

我们可以把  $X$  上的一个圈在内部画线, 使得它们分成  $A_\alpha$  上的一些小圈, 通过在分割线上“折返”, 将大圈路径分成两个小圈的拼接. 由于在自由基的语境下, 折返是在两个不同空间上进行的, 但是对于  $X$  确实一个空间, 所以我们要在自由积的语境下把折返的元素  $i_{\alpha\beta}(\omega)i_{\beta\alpha}(\omega)^{-1}$  模掉.

### Exercise 1.4.2

Let  $X_\alpha$  be a collection of path-connected spaces, we consider the wedge sum  $\vee_{\alpha} X_\alpha$  with the basepoints  $x_\alpha \in X_\alpha$ . If each  $x_\alpha$  is a deformation retract of an open subset  $U_\alpha$  of  $X_\alpha$ , then

$$\pi_1(\vee_{\alpha} X_\alpha) \simeq *_{\alpha} \pi_1(X_\alpha)$$

*Proof.* We have  $X_\alpha$  is a deformation retract of its open neighborhood  $A_\alpha = X_\alpha \vee_{\beta \neq \alpha} U_\beta$ . The intersection of two of more  $A_\alpha$ 's is  $\vee_{\alpha} U_\alpha$ , which is a deformation retract to the basepoint. Then the conclusion follows from Van Kampen's Theorem.  $\square$

## 1.5 Covering Spaces

### Definition 1.5.1

Let  $p : \bar{X} \rightarrow X$  be a surjective map. We say an open subset  $V$  of  $X$  is

*evenly covered* by  $p$ , if  $p^{-1}(V)$  is a disjoint union of open subsets of  $\bar{X}$ :

$$p^{-1}(V) = \coprod_i U_i$$

where, each  $U_i$  is mapped homeomorphically onto  $V$  by  $p$ . In this case, we shall refer to each  $U_i$  as a *sheet for  $p$  over  $V$* . If the space  $X$  can be covered by open subsets which are evenly covered by  $p$ , then we say that,  $p$  is a *covering projection*; the space  $\bar{X}$  is called a *covering space* of  $X$ .

## 1.6 Exercises